GD, SGD and Polyak stepsizes for logistic regression on the California housing data

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Idea

After the linear regression, I repeat the same experiments with a model that cannot be solved exactly: the logistic regression. Of course, the solution can be found with some approximations, which I used as well, but I wanted to use another model without overcomplicating things and to concentrate on GD, so I use this one. There is no formula for the solution $w^\star$, so we compute it once with Newton's method, to machine precision. GD and SGD do not use it; it is only the reference for the error $\|w^k - w^\star\|^2$ and gives $f^\star$ for the Polyak stepsize.

Data and model

We take the same California house prices data and the same features as before (the logarithm of some of the features, and the outliers clipped to $[-3, 3]$), but now the task is a classification: a district is "expensive" ($y_i = +1$) if its median house value is above the median of the training set, and $y_i = -1$ otherwise. We minimize the logistic loss with a small ℓ_2 term,

$$f(w) = \frac{1}{n}\sum_{i=1}^{n} f_i(w), \qquad f_i(w) = \log\bigl(1 + e^{-y_i\langle x_i, w\rangle}\bigr) + \frac{\lambda}{2}\|w\|^2, \qquad \lambda = 0.01.$$

The ℓ_2 term is needed to make f strongly convex.

The constants. With $z_i = y_i\,\langle x_i, w\rangle$ and the sigmoid $\sigma(t) = \frac{1}{1+e^{-t}}$, the gradient and the Hessian of f_i are

$$\nabla f_i(w) = -y_i\,\sigma(-z_i)\,x_i + \lambda w, \qquad \nabla^2 f_i(w) = \sigma(z_i)\bigl(1-\sigma(z_i)\bigr)\,x_i x_i^\top + \lambda I.$$

With $p = \sigma(z_i) \in (0,1)$ we have $p(1-p) = \frac{1}{4} - \bigl(p - \frac{1}{2}\bigr)^2 \le \frac{1}{4}$, with equality only for $z_i = 0$, i.e. when the model is completely unsure about the point x_i.

- L_i: the matrix $x_i x_i^\top$ has the eigenvalue $\|x_i\|^2$ in the direction x_i and 0 in all the other directions, so the largest eigenvalue of $\nabla^2 f_i(w)$ is at most $\frac{\|x_i\|^2}{4} + \lambda$. It is attained when $z_i = 0$ (for example at $w = 0$), hence $L_i = \frac{\|x_i\|^2}{4} + \lambda$.
- L: the Hessian of f is the average $\frac{1}{n}\sum_i \sigma(z_i)\bigl(1-\sigma(z_i)\bigr)x_i x_i^\top + \lambda I$. Bounding every factor by $\frac{1}{4}$ gives $L \le \lambda_{\max}\bigl(\frac{1}{4n}X^\top X\bigr) + \lambda$, and at $w = 0$ all the factors are exactly $\frac{1}{4}$, so $L = \lambda_{\max}\bigl(\frac{1}{4n}X^\top X\bigr) + \lambda$. As for the linear regression, L is smaller than the average $\bar{L} = \frac{1}{n}\sum_i L_i$, because the directions x_i are different.
- μ: the factor $\sigma(1-\sigma)$ is positive, but it tends to 0 far from the data (when the model is very confident), so the only lower bound for all w is $\nabla^2 f(w) \succeq \lambda I$, hence $\mu = \lambda$. Without the ℓ_2 term we would have $\mu = 0$.

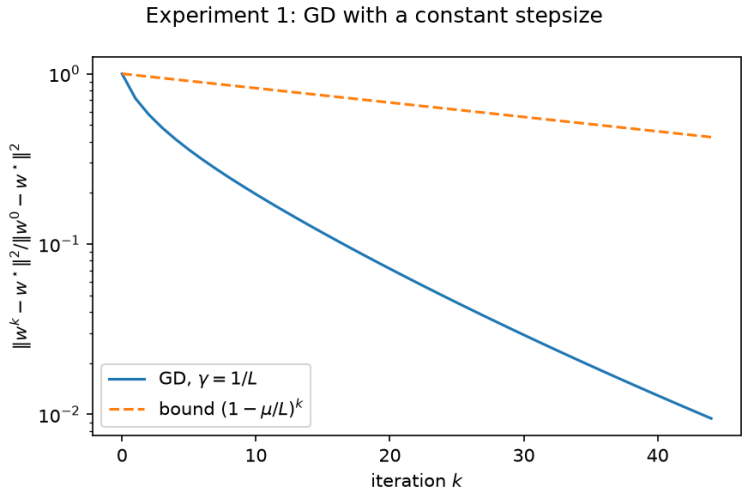
For our data $\mu = 0.01$, $L = 0.519$, $\bar{L} = 1.98$ and $L_{\max} = 11.34$, so $A = 2L_{\max} = 22.7$ for SGD-US and $A_{\mathrm{IS}} = 2\bar{L} = 3.96$ for SGD-IS. These are worst-case constants.

All methods use one target: $\|w^k - w^\star\|^2 \le 10^{-2}\,\|w^0 - w^\star\|^2$.

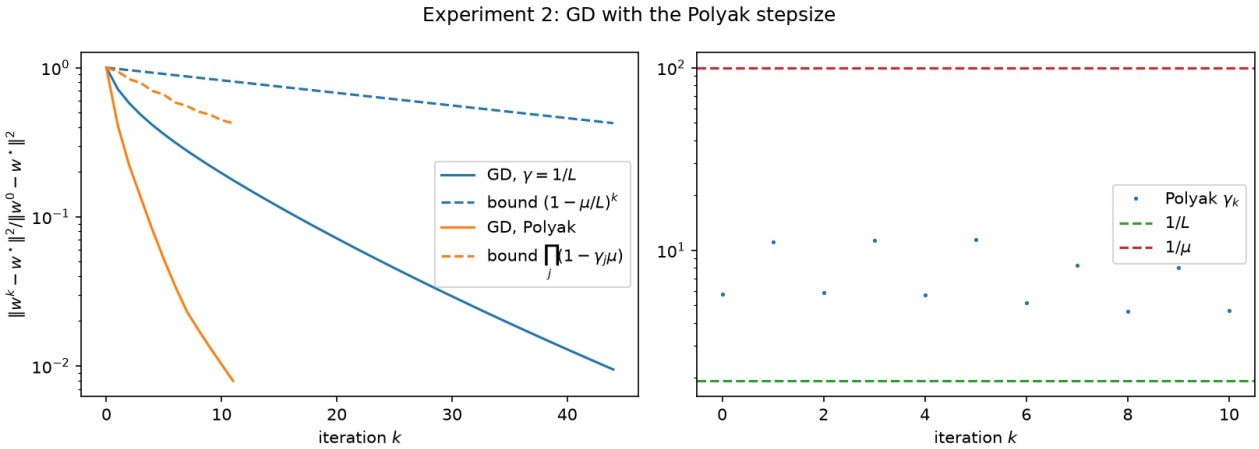
The code is the file `Homework/Exercises/hw_2026_11_11/sgd_logistic.ipynb`.

Experiments

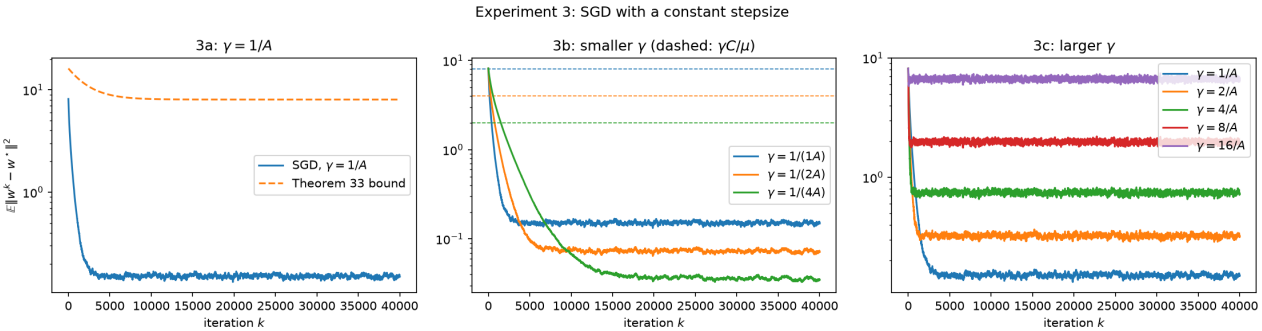
The first experiment is GD with the constant stepsize $\frac{1}{L}$.

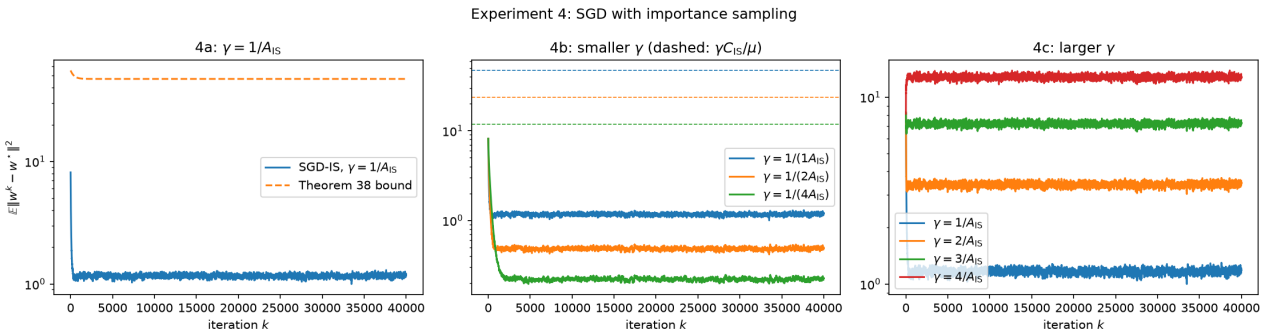


The second one is GD with the Polyak stepsize, again with the theoretical bounds and the comparison with the simple GD. Here the Polyak stepsize helps even more than for the linear regression, because it adapts to the actual curvature, which is much smaller near $w^\star$ than the worst case L.



After them come SGD-US and SGD-IS with their theoretical maximal stepsizes $\frac{1}{A}$ and with other stepsizes. Unlike the linear regression, a much larger γ than allowed does not make SGD diverge, since the gradients of the logistic loss are bounded; the method only settles in a much larger neighbourhood.





Comparison

The last experiment is the comparison of all the methods, both by the number of iterations and by the time.

